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Lab #2

Digital Meters (DMM)

Introduction

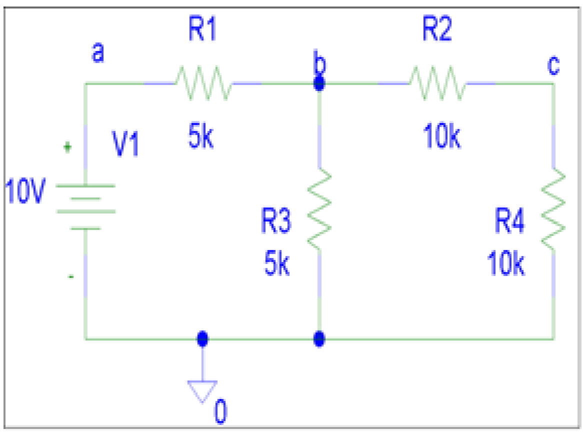
The objective of this laboratory was to understand the measurement functionality and practical behavior of digital multimeters (DMMs). Throughout this lab, it is shown that, properties of parallel and series circuits are tools of manipulation to maintain key conditions for the meter. By adding resistors, we can control the new RM (resistance of the meter) to not exceed the IFS and to control the voltage going into the meter. By changing the scaling resistor in series, we can change the voltage into the meter. To control the current in the circuit, we can add a resistor in parallel. Most critically, the full scale current of the meter (IFS) cannot be exceeded.

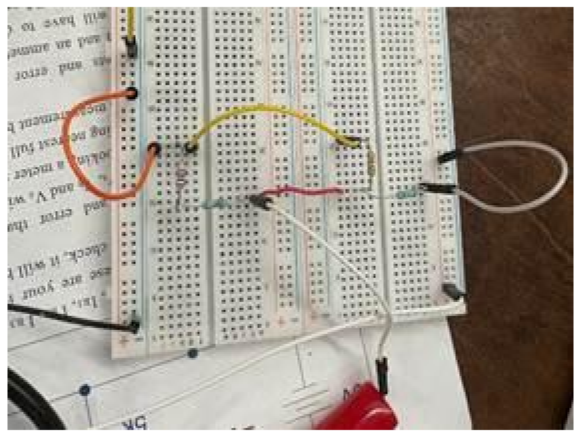
Additionally, multimeters possess an internal meter resistance that affects both circuit behavior and the accuracy of measured values. The full-scale current is a key factor in determining a meter's sensitivity, as a smaller one allows the measurement of smaller signals. Although analog meters are used less frequently today than digital meters many of the same underlying principles still apply. Digital multimeters, generally more applicable, rely heavily on current-to-voltage conversion techniques to perform measurements. In some cases in the modern world, analog meters remain useful due to their visual indication of trends and changes.

Overall, this laboratory explored how DMMs interact with circuits when used as voltmeters, ammeters, and ohmmeters, thus highlighting both their advantages and their non-ideal effects on measured systems.

Part 1. DMM DC Voltage Measurements

1. In order to see the effect of the DMM voltmeter on a circuit, we built a simple voltage divider as seen below in the schematic in order to measure different V_{Out} cases.





IMG 1. Schematic of the voltage divide. **IMG 2.** Physical circuit based on figure 1.

Theoretical Resistances (Ω)	**Exact Values measure w/ Multimeter (Ω)**
10M	9.83M
5M	5.025M
1M	0.997M
100k	98.75k

10k	9.8k
1k	0.99k

Table 1.

While R3 = 5MΩ and R4 = 10MΩ remained constant throughout our built circuit, we changed R1 and R2 with different resistor values to measure and view the effect it had on V_{Out}.

Resistor 1 (Ω)	**Resistor 2 (Ω)**	**V_{Out} (V)**
5M	10M	3.59
1M	1M	7.17
100k	100k	9.59
10k	10k	9.93
1k	1k	9.96

Table 2.

<u>Question:</u> At what level of circuit resistance does the meter have less than 1% effect on the "True Value"?

Based on our data above, the meter has less than 1% effect once the circuit resistance is 10kΩ and lower. This is because a 1% meter effect essentially means the reading stays at $9.90V \leq V_{measure} \leq 10.10V$. At R1 = R2 = 100kΩ, V_{Out} = 9.59 which is roughly 4.1%. At R1 = R2 = 10kΩ, V_{Out} = 9.93V which is roughly 0.7% and then at R1 = R2 = 1kΩ, V_{Out} = 9.96v which is about 0.4%.

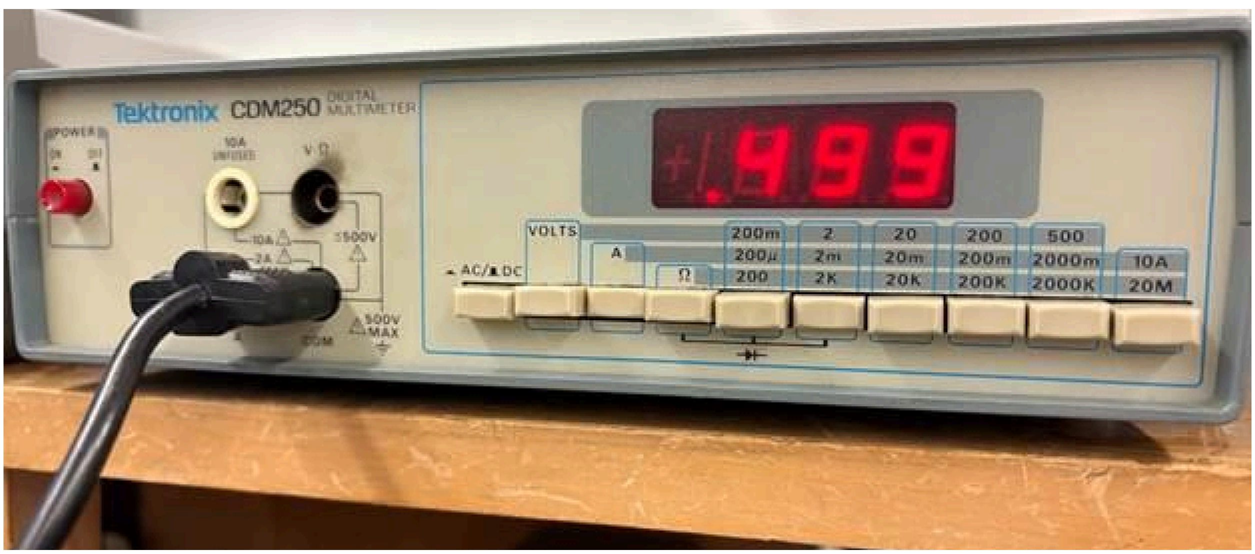
Part 2. DMM DC Ammeter Measurement

In order to see the effect of the DMM Ammeter in a circuit, we built a simple series circuit with a power supply, resistor and ammeter as shown in figure 3 below. A 2nd DMM voltmeter was used to measure the voltage across the original ammeter called the Burder voltage. The Burden voltage is the amount of voltage the meter requires to make a measurement and thus is directly related to the ammeter R_M value as Vammeter (burden voltage) over the amperage value is R_M.

Theoretical Resistance (Ω)	Measured Resistance (Ω)
5k	4.7k

Table 3.

We measured the current at each available different setting for the Tektronix CDM250 digital multimeter and measured the Burden voltage each time the setting was switched.



IMG 4. Tektronix CDM250 Digital Multimeter

Current CDM250 Setting	Measured Current	Measured Burden Voltage
200µA	160.8µA	0.161 V

2mA	0.189mA	19.18mV
20mA	0.19mA	2.137mV
200mA	0.001mA	0.394mV
2000mA	0.0	0.218mV

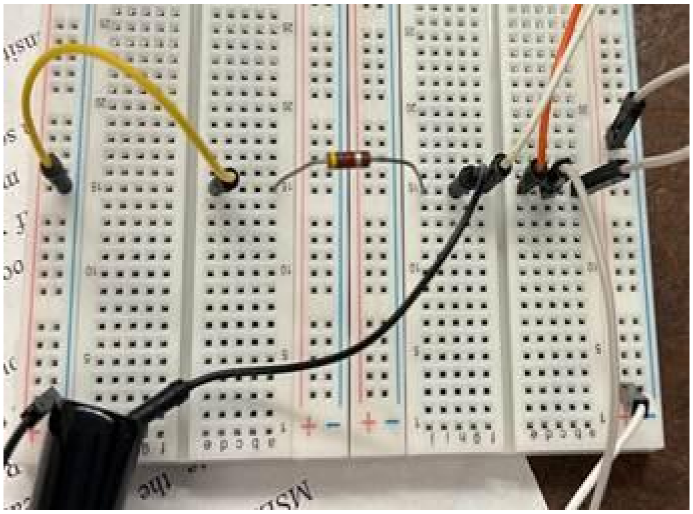
Table 3.

Part 3. Ohmmeter (2 Wire) Measurements

DMM ohmmeters work by sending out a reference current and measuring the voltage across the unit to determine the unknown R value by $V_{Measured}$ divided by I_{Source}. This is a basic example of a SMU (Source Measurement Unit) type of measurement.

Figure 5 and 6. Diagram of a (2 wire) ohmmeter and schematic for testing.

1. Determine the reference current for ohmmeter on the kΩ setting by connecting a resistor under test and ammeter in series with the ohmmeter.



IMG 7. Series tester circuit

Image 8. Keysight DMM Ohmmeter resistance reading (2 wire)

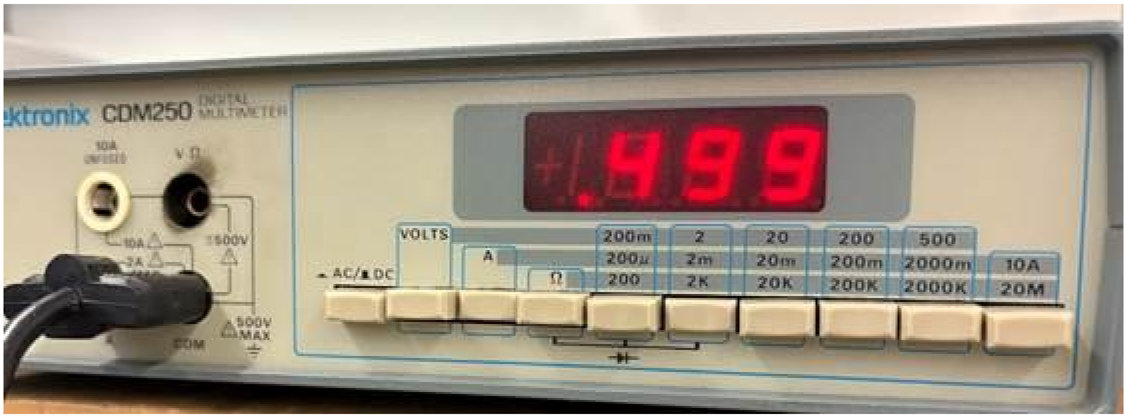


Image 9. Amperage reading using CDM250 DMM, 0.499mA

Using the Tektronix CDM250 DMM, our current reading was 0.499mA.

Using the Keysight DMM Ohmmeter, our resistance was 0.10074kΩ.

Conclusion

This experiment showed that measurement devices are not electrically neutral and that they can actually unintentionally alter the circuits they are measuring and connected to. The internal resistance of a meter plays a key role in determining both circuit operation and the accuracy of the measured result. Analog meters, in particular, tend to introduce noticeable loading effects while current-measuring modes in both analog and digital instruments can significantly affect circuit conditions. Although digital multimeters greatly minimize interference during voltage measurements through their high input impedance (especially as opposed to traditional analog metrs), their current ranges still influence circuit behavior in a manner similar to traditional analog meters. Overall, this lab improved our understanding of how different types of multimeters interact with circuits and emphasized the importance of accounting for meter characteristics when making electrical measurements.